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Unit – 3A → Air Pollution

Introduction to Pollution

Introduction:

- Environmental pollution means “the addition of any substance or any form of energy into the environment at a rate faster than it can be dispersed/decomposed in a harmless form.”
- Any change which adversely affects living organisms is known as pollution.
- Pollution may be natural (due to natural calamity) or man-made (human activity).
- Substance that causes pollution is known as a “**pollutant**”.
- Not all pollutants are harmful, if not present in excess concentration.

Types of pollution:

- If any of the components of the environment gets affected, subsequently living organisms are also affected, thus pollution occurs.
- As discussed, pollution can be natural (volcanic eruption, UV radiation, forest fire) or man-made.
- In the beginning, nature was able to absorb harmful substances, but after industrialization, overpopulation, and urbanization nature was unable to process or neutralize harmful by-products of human activity.
- Pollution in various components can be classified as;
 - (1) Air pollution
 - (2) Water pollution
 - (3) Land pollution
 - (4) Noise pollution
 - (5) Thermal pollution

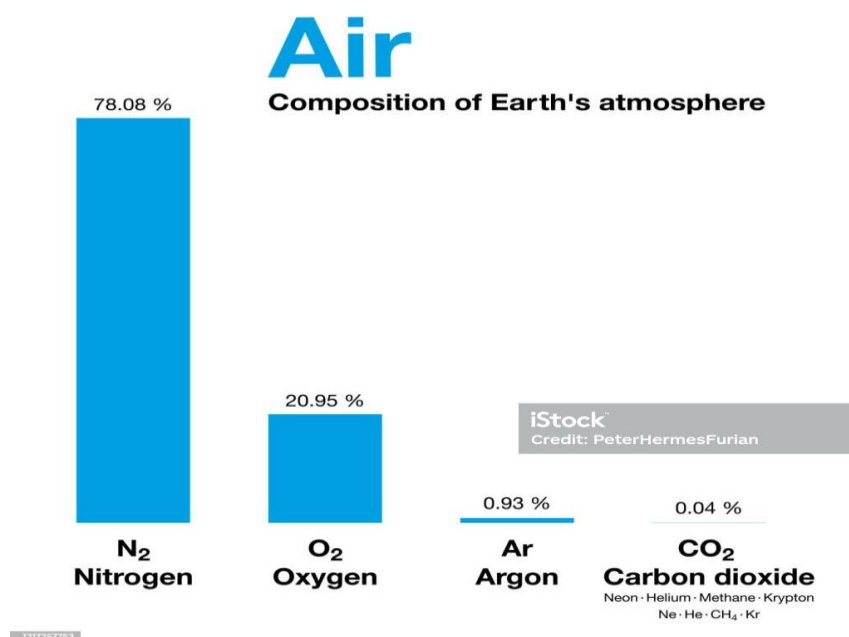
Air Pollution

Introduction:

- Air constitutes 80% of the man's daily intake of material by weight. We breathe about 22000 times a day on an average, inhaling 16 kg of air a day. This shows how importance fresh and unpolluted air.
- The effects of other type of pollution like water, land and noise pollution are limited to local area only where pollutants are discharged.
- But effect of air pollution is both local as well as global in nature so it is very important to learn about air pollution and control it.
- Some of the global effect of air pollution are global warming, acid rain, thinning of ozone layer.
- "Air pollution is the presence in ambient atmosphere of substances, generally resulting from human activity in sufficient concentration, present for sufficient time and causes harmful effect on humans, plants and animals."

Composition of air:

- Atmosphere is mixture of various gases, water vapor and particulate matter.
- The major gases in a pollution-free dry environment are nitrogen (78%), Oxygen (21%), Argon (0.9 - 1%) and Carbon dioxide.
- Minor gases include Neon, Helium, Hydrogen, Methane, Ozone.



Structure of atmosphere:

→ Based on the temperature profile and other related phenomena, the atmosphere is divided into five major layers.

1. Troposphere;

- is the lower portion of the atmosphere which extends up to 8 Km at the poles and 16 Km at the equator on average it extends up to 10-12 km of the earth's surface. The temperature in this region decreases at the rate of 5-7 c/km there are strong vertical air movements in this region which are responsible for the rapid and complete mixing.

2. Stratosphere;

- It is above the troposphere and extends up to 50-55 km up to about 20 Km temperature remains constant and then increases with an increase in height. The ozone layer is in this layer.

3. Mesosphere;

- It is the layer above the stratosphere and extends up to 80 km. here temperature decreases slowly with altitude but then sharply to about -75 °C.

4. Thermosphere;

1. Thermosphere extends from 80 km to 550 km. Thermosphere is divided into two parts: the upper part and the lower part. The upper part is known as the exosphere and the lower part is known as the ionosphere. The ionosphere helps in radio communication and exosphere is used as an orbit for satellites.

Sources of Air pollution

→ Sources of air pollution can be mainly classified in to:

1. Natural
2. Man-made or anthropogenic

Natural sources:

1. Volcanic eruption
2. Marshy gases from marsh places
3. Forest fire
4. Dust storm
5. Bacteria and other organism
6. Photochemical smog
7. Pollen grains
8. Spores
9. Salt spray from ocean

Man-made sources:

1. Industrial units
2. Thermal power plants
3. Automobile exhaust
4. Fossil fuel burning
5. Agricultural activity
6. Mining
7. Nuclear explosion
8. Domestic wood burning
9. Air crafts

→ Man-made sources can be generally classified in to:

1. Point or stationary sources: sources which adds pollutants into atmosphere at particular points, likes a chimney.

2. Mobile or line sources: these types of sources are automobile as these adds pollutants along a narrow belt and over a long distance, while is motion.
3. Area sources: towns & cities add smoke and gases over a wide area and so it is called area sources of air pollution.

Classification of Air pollutants

→ Air pollutants can be classified as per the following criteria:

- (1) As per the origin of air pollutants
- (2) As per the state of matter of the pollutants
- (3) As per the Presence in the environment

Classification based on Origin of pollutants:

→ Based on the origin air pollutants can be;

1. Primary pollutants

- They are emitted directly from the sources and are found in the atmosphere in the form in which they were emitted.
- Sulphur oxides, nitrogen oxides, hydrocarbon, ash, etc.

2. Secondary pollutants

- They are formed in the atmosphere by chemical interaction between primary pollutants and atmospheric compounds.
- Ozone, sulphur trioxide, PAN (peroxyacyl nitrate), ketones, aldehydes, etc.

Classification based on State of matter of pollutants:

→ Based on the state of matter air pollutants can be;

1. Gaseous pollutants

- These pollutants are in gaseous state at normal pressure and temperature condition.
- CO, CO₂, NO_x, SO_x, etc.

2. Particulate air pollutants

- These include suspended droplets, solid particles or their mixture in the atmosphere.
- Aerosols, dust, mist, fumes, etc.

Classification based on presence in environment:

→ Based on presence in environment air pollutants can be;

1. Indoor pollutants

- These are the pollutants which can be observed in an indoor environment.
- Cleaning agent, mosquito repellent, cigarette smoke, etc.

2. Outdoor pollutants

- These are the pollutants which can be observed in an outdoor environment.
- Automobile exhaust, industry exhaust, etc.

Common air pollutants – Sources and effects

→ Common air pollutants include;

1. CO: Colourless, odourless, tasteless gas. Chemically inert gas under normal condition of temperature and pressure.

- Sources of CO

- Natural processes – volcanoes, natural gas emission, electrical charge during cloud forming, marsh gases production adds CO in small amount.
- Transportation sources contribute about 64% of CO in air.
- Forest fire and agricultural burning contributes about 17% of CO in atmosphere.
- Industrial processes such as electric furnace and blast furnace in iron and steel industry, petroleum refining, etc, causes 9.6% of CO in air.

- Effects of CO

- It reduces oxygen carrying capacity of blood by selectively combining with hemoglobin forming carboxy hemoglobin, this causes giddiness, laziness and exhaustion.
- It reduces vision and causes cardiovascular diseases.
- CO is very dangerous asphyxiant and its high levels can be fatal to human life.

2. CO₂: Content of CO₂ has increased 15% in the last century.

- Sources of CO₂

- Fossil fuel burning
- Jet plane uses O₂ and releases CO₂
- Deforestation
- Forestry

- Effects of CO₂

- CO₂ is comparatively less dangerous than CO and causes headache and nausea.
- Its major effect is on climate, it is green house gas, leading to global warming, and subsequently resulting in melting of glacier.

3. NO_x : NO_x group can have eight possible oxides of nitrogen, but only NO, NO_2 and N_2O are considered as pollutants.

- Sources of NO_x

- Fuel combustion in automobiles and industries
- Lightening
- Forest fire
- Bacterial decomposition of organic matter
- Natural ionizing adiation

- Effects of NO_x

- Like CO, nitric oxide (NO) can also combine with hemoglobin and reduce oxygen carrying capacity of blood.
- NO is moderately toxic as its concentration in air is not high.
- NO_2 is more toxic, it irritates alveoli of lung and its high concentration may cause acute bronchitis.

4. SO_x : it includes SO_2 and SO_3 . SO_2 is colourless gas having pungent and suffocating odour.

- Sources of SO_x

- Most of the SO_x pollution (67%) is due to volcanic activities and other natural sources.
- Rest of the SO_x is due to man-made activities like
 - Burning of solid and fossil fuels.
 - Transportation
 - Industries – paper, refineries, sulfuric acid plants

- Effects of SO_x

- These tends to irritate the mucous membranes of respiratory tracks
- High concentration of oxides of sulphur can cause Bronchitis.
- Marble and mortar statues are affected.
- Clothes, leathers, and papers are also affected.
- Plants are sensitive to high concentration of SO_2 suffer chlorosis, metabolic inhibition, plasmolysis and even death.

5. Particulate air pollutant:

- These are small particles of liquid and solid present in the atmosphere in fairly large numbers and sometimes pose serious air pollution problems.
- PM sizes ranges from 0.02 micron to 500 micron.
- According to size and mode of formation it can be classified as;
 1. Aerosols: these include all air borne suspension of solid or liquid particle smaller than 1 mm. For example: Dust, smoke, mist and fumes.
 2. Dust: these are small weathered earth particles, formed by breaking of larger masses of rock and soil either by natural disintegration or by mechanical disintegration.
 3. Smoke: it consists of fine solid particles resulting from incomplete combustion of organic matter like coal, wood, tobacco and other chemical processes.
 4. Fumes: these are the fine solid particles formed by condensation of vapour of solid material. They are released from chemical metallurgy processes.
 5. Mist: it consists of liquid droplets formed by condensation of liquid vapour.
 6. Fog: if the mist is made up of water vapour and it is in high concentration, enough to reduce visibility, it is known as fog.
 7. Flyash: this consists of finely divided non-combustible particles present in gases arising from fuel combustion. it contains inorganic metallic or mineral substances.
 8. Soot: incomplete combustion of carbon containing materials release carbon particles. A collection of such particles is known as soot.
- Sources of PM:
 - Numerous natural activities like volcanic eruption, blowing of dust by wind, spraying of salt and other solid particles by ocean and sea.
 - Anthropogenic activities release, fly ash from power plant, melting and mining operations, and smoke from incomplete combustion.
- Effects of PM:
 - PM of size less than 10 micron can enter in lungs through respiration and may cause irritation, aggravated asthma, chronic bronchitis, etc.

- PM of lead, cadmium, nickel can be hazardous to health.
- It accelerates corrosion of metals, damages buildings, sculptures, paints, etc.
- Aerosols releases of emission from jet and supersonic planes deplete the ozone layers.
- Dust coating on leaves affects photosynthesis and reduces plant growth.
- PM reduces visibility, has influence on the climate through formation of cloud, rain and snow by acting as a nucleus upon which water condenses.

Effects of Air pollution

Effect of air pollution on human:

- Eye irritation can be caused by many air pollutants such as NO_x, O₃, PAN, smog, PM.
- Nose and throat irritation can be caused by SO₂, NO_x, pesticides.
- Gaseous pollutants can cause odour nuisance even at low concentration.
- Irritation of respiratory tract can be caused by SO₂, NO_x, CO, etc.
- Increase in mortality and morbidity rate.
- Pollens can initiate asthma attacks.
- High concentration of SO₂, NO₂, SPM can cause bronchitis and asthma.
- CO and NO can combine with hemoglobin and reduces oxygen carrying capacity of the blood, and consequently increases stress on those suffering from cardiovascular and pulmonary diseases.

Effects of air pollution on vegetation

- Bleaching of leaves, necrosis (killing of tissue)
- Premature ageing, suppressed growth, etc.

Effects of air pollution on property/material

- SO₂ in presence of O₂ and moisture can lead to acid rain, and effect protective coating or building material.
- O₃ and PAN may cause cracking in rubber tyres and various form of electrical insulation.

Control of Air pollution

→ Air pollution cannot be fully abated but can be controlled if certain measures are taken.

→ Controlling measures for air pollution can be mainly divided into two groups;

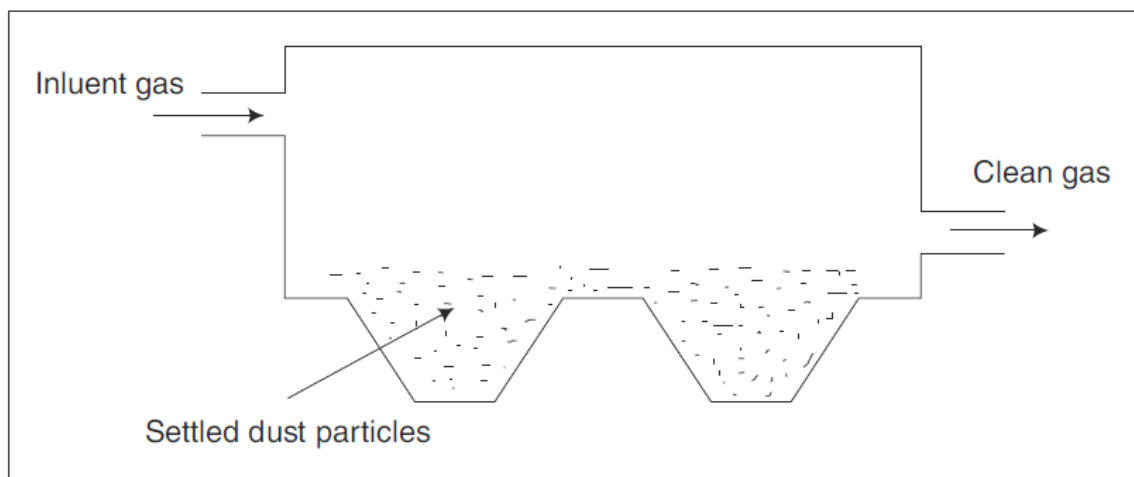
1. Preventive measures: measures aimed right at the source to enable emission in lower concentration.
2. Control measures: measure taken, when prevention is not possible, to reduce the polluting power of the pollutant.

Preventive measures:

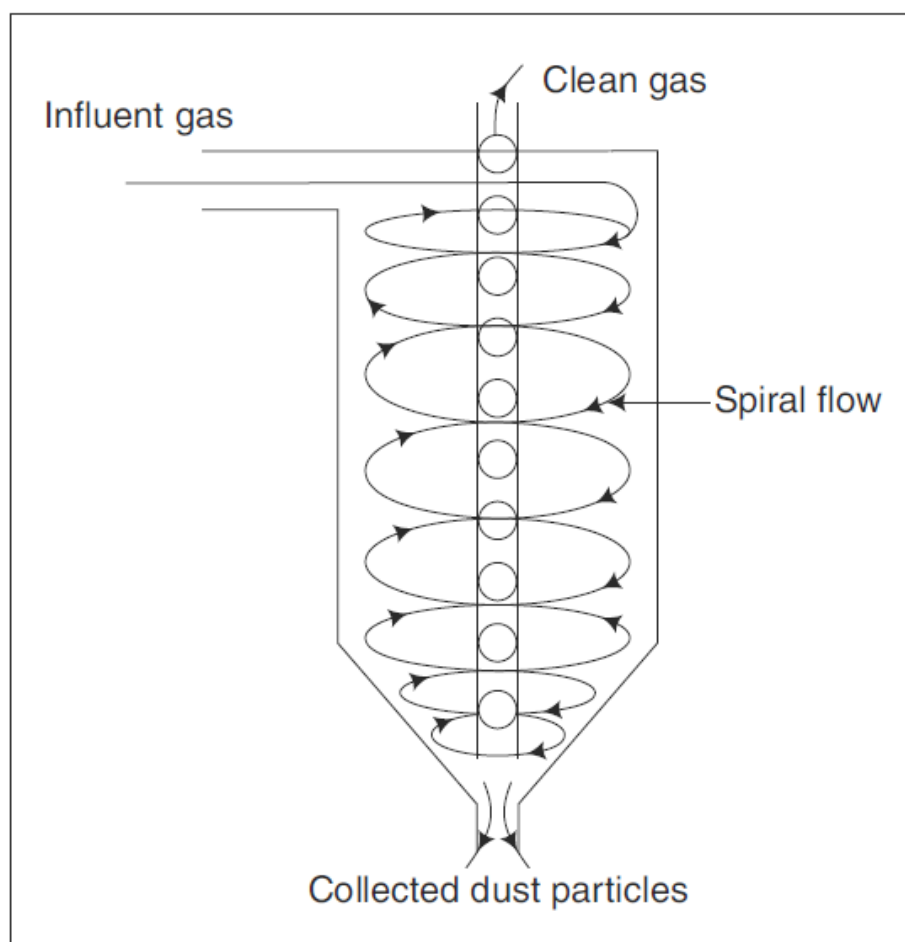
1. Selection of suitable fuel: Selection of suitable fuel;
 - Low sulphur coal in power plant
 - Using natural gas in place of coal for power generation
 - Using LPG/CNG instead of petrol or diesel in automobiles
2. Modification in industrial process and/or equipment to reduce the emission
 - If coal is washed before pulverization, then fly-ash emission is reduced considerably
3. Select suitable site and zoning of industrial unit
4. By using proper environmental impact study before installing the industry.

Control measures:

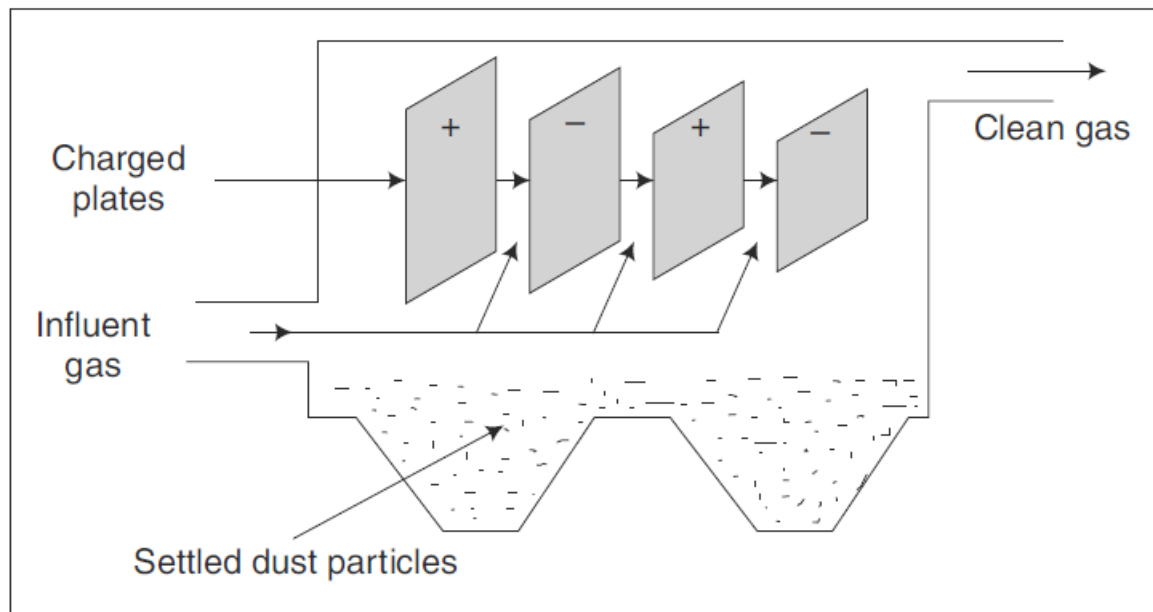
- Changing pollutants to a less toxic form
- Release through tall chimney for greater dispersion
- Destroying the pollutants by thermal or catalytic combustion
- Collecting the pollutants by using equipment to prevent escape into the atmosphere
1. Cyclone separator
 2. Gravity settler
 3. Scrubber
 4. Electrostatic precipitator



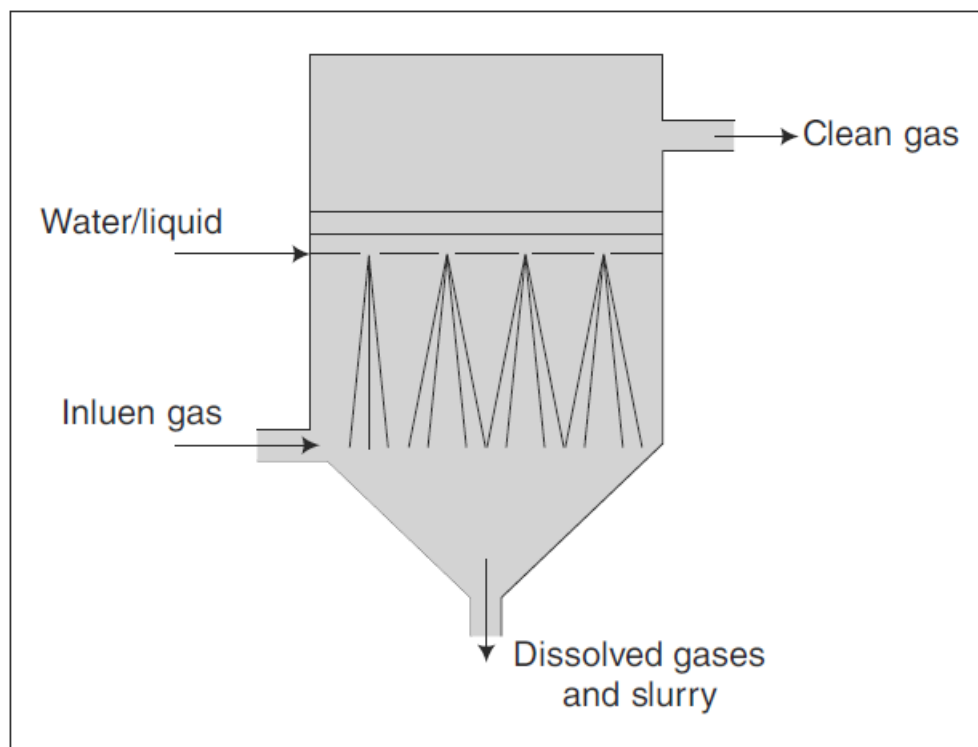
Gravity settler



Cyclone separators



Electrostatic Precipitator



Scrubbers